

Introduction

The **standard error** of a statistic is the standard deviation of its sampling distribution. It quantifies how much the statistic would vary across hypothetical repetitions of the sampling process. Every confidence interval and every t- or z-statistic is built on an estimate of the standard error, so understanding and correctly computing SEs is central to competent data analysis.

Prerequisites

A reader should know what the sampling distribution is, what a sample standard deviation is, and how to use the `sd()` and `mean()` functions in R.

Theory

For an iid sample $X_1, \dots, X_n$ from a population with variance σ^2 , the sample mean $\bar{X}$ has variance σ^2/n and therefore

$$\text{SE}(\bar{X}) = \sigma/\sqrt{n}.$$

Because σ is usually unknown, it is estimated by the sample SD s , giving the plug-in estimator $\widehat{\text{SE}}(\bar{X}) = s/\sqrt{n}$.

For a binary proportion $\hat{p} = k/n$, the standard error under binomial sampling is

$$\text{SE}(\hat{p}) = \sqrt{p(1-p)/n}, \quad \widehat{\text{SE}}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}.$$

For a difference in means across two independent samples with variances s_1^2, s_2^2 ,

$$\widehat{\text{SE}}(\bar{X}_1 - \bar{X}_2) = \sqrt{s_1^2/n_1 + s_2^2/n_2}.$$

For regression coefficients, standard errors come from the diagonal of $\hat{\sigma}^2(X^\top X)^{-1}$; R returns them in `summary(fit)$coefficients`.

For statistics without a closed-form SE (ratios, quantiles, custom summaries), the **bootstrap** gives a general-purpose estimator: resample with replacement B times, compute the statistic for each resample, and use the SD of the B bootstrap statistics as $\widehat{\text{SE}}$.

Assumptions

The formula-based SEs above assume iid sampling and, for the finite-sample versions, enough regularity for the CLT to justify using them in normal-approximation CIs. The bootstrap SE is consistent under weaker conditions but inherits the design: it is accurate only when resampling units are the same as the independent sampling units (so bootstrap at the cluster level for cluster samples).

R Implementation

```
library(boot)

set.seed(2026)
x <- rnorm(60, mean = 75, sd = 12)

se_mean <- sd(x) / sqrt(length(x))
se_mean
```

```

y <- c(rep(1, 24), rep(0, 76))
p_hat <- mean(y)
se_prop <- sqrt(p_hat * (1 - p_hat) / length(y))
se_prop

iqr_stat <- function(d, i) IQR(d[i])
b <- boot(x, statistic = iqr_stat, R = 5000)
se_iqr <- sd(b$t)
se_iqr

```

The first two blocks use the closed-form SEs for the mean and a proportion; the third uses the bootstrap for the IQR, which has no convenient closed form.

Output & Results

```

[1] 1.552      # SE(mean)
[1] 0.0427     # SE(proportion)
[1] 2.11      # bootstrap SE(IQR)

```

Each of these is the estimated typical distance between the sample statistic and the population parameter it estimates.

Interpretation

The standard error is what gets reported alongside the estimate: “mean 75.8 (SE 1.6)” or “proportion 0.24 (95% CI 0.16–0.32)”, where the CI is approximately the estimate plus/minus 1.96 times the SE. A reviewer glancing at the SE can judge whether the study is precise enough to support its claims.

Practical Tips

- Always distinguish the SD (spread of individual values) from the SE (spread of a statistic); papers regularly report the wrong one.
- For proportions near 0 or 1, the normal-approximation SE is poor; prefer Wilson or Clopper-Pearson intervals, which are available via `binom.test()` and the `binom` package.
- For differences between groups, the correct SE is the SE of the difference, not either group’s SE by itself.
- When you lack a closed-form SE, reach for the bootstrap rather than inventing one; `boot::boot()` handles the resampling machinery robustly.
- Do not report an SE without stating what statistic it refers to – “SE = 1.6” alone is ambiguous; “SE of the mean = 1.6” is clear.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain

- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr